

Final Example of Lec 3 $\Rightarrow \exp\left(\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} t\right) = \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}.$

Here is another approach, and for $\begin{bmatrix} a & b \\ -b & a \end{bmatrix} t$ more generally

Define $J := \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$; note that $J^2 = -I$.

Define $F: \mathbb{C} \rightarrow \mathbb{R}^{2 \times 2}$ by $F(a+ib) = aI + bJ \quad \forall a, b \in \mathbb{R}.$

Clearly F is $\mathbb{R}$ -linear (and hence continuous). Note that, since $J^2 = -I$,

$$\begin{aligned} F((a+ib)(c+id)) &= F(ac-bd + i(bc+ad)) = (ac-bd)I + (bc+ad)J \\ &= (aI + bJ)(cI + dJ) = F(a+ib)F(c+id) \end{aligned}$$

Thus, $F(z_1 z_2) = F(z_1) F(z_2) \quad \forall z_1, z_2 \in \mathbb{C}$

Using $\odot$, $\otimes$, we deduce that

$$F\left(1 + z + \frac{z^2}{2!} + \dots + \frac{z^k}{k!}\right) = I + F(z) + \frac{F(z)^2}{2!} + \dots + \frac{F(z)^k}{k!}$$

$\forall z \in \mathbb{C}, k \in \mathbb{N}.$

Since F is continuous (by $\odot$), taking limits as $k \rightarrow \infty$ above yields

$$F(e^z) = e^{F(z)} \quad \forall z \in \mathbb{C}.$$

Writing $z = a + ib$ for $a, b \in \mathbb{R}$, Euler's formula yields $e^z = e^a e^{ib} = e^a (\cos b + i \sin b)$, so the above identity yields

$$\begin{aligned} e^a (\cos b \, I + \sin b \, J) &= F(e^{a+ib}) = e^{F(a+ib)} = e^{aI + bJ} \\ &= \exp\left(\begin{bmatrix} a & b \\ -b & a \end{bmatrix}\right). \end{aligned}$$

Thus,

$$\exp\left(\begin{bmatrix} a & b \\ -b & a \end{bmatrix} t\right) = e^{at} (\cos(bt) I + \sin(bt) J) \\ = e^{at} \begin{bmatrix} \cos(bt) & \sin(bt) \\ -\sin(bt) & \cos(bt) \end{bmatrix}$$

In particular, $\exp\left(\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} t\right) = \begin{bmatrix} \cos(bt) & \sin(bt) \\ -\sin(bt) & \cos(bt) \end{bmatrix}.$

Rmk (abstract algebra) properties $\circledast$, $\circledstar$ above show that $F: \mathbb{C} \rightarrow \mathbb{R}^{2 \times 2}$ is a ring monomorphism; it is a ring isomorphism from $\mathbb{C}$ onto the ring of all 2×2 real matrices of the form $\begin{bmatrix} a & b \\ -b & a \end{bmatrix}.$

Exercise show that $\exp\left(\begin{bmatrix} a & b \\ -b & a \end{bmatrix} t\right) = e^{at} \begin{bmatrix} \cos(bt) & \sin(bt) \\ -\sin(bt) & \cos(bt) \end{bmatrix}$ via the following alternative approach. Write $\begin{bmatrix} a & b \\ -b & a \end{bmatrix} t = at I + bt J$ w/ J as above. Note that $[at I, bt J] = 0$. Thus, $e^{(at I + bt J)} = e^{at} e^{bt J}.$

Finally, complete the derivation by using $J^2 = -I$ to compute $e^{bt}J$ from the definition, using the Maclaurin series for $\sin, \cos$.

Real normal form of $A \in \mathbb{R}^{n \times n}$

2x2 case first

Suppose $A \in \mathbb{R}^{2 \times 2}$ has a complex conjugate pair of eigenvalues $\lambda = a + ib$, $\bar{\lambda} = a - ib$ w/ $a, b \in \mathbb{R}$, $b \neq 0$.

Let $v = u + iw$ be an e-vector for λ , w/ $u, w \in \mathbb{R}^2$.

To find an e-vector for $\bar{\lambda}$, take conjugate of $Av = \lambda v$ to obtain $A\bar{v} = \bar{\lambda}\bar{v}$, so $\bar{v}$ is an e-vector for $\bar{\lambda}$.

Since $\lambda \neq \bar{\lambda}$, $\{v, \bar{v}\}$ is linearly indep (since e-vectors from distinct eigenspaces).
(over $\mathbb{C}$)

Cor $\{u, w\}$ is lin. indep.

Pf Suppose not. Then either $u = \alpha w$ or $w = \beta u$ for some $\alpha, \beta \in \mathbb{R}$.

If $u = \alpha w$, then $v = u + iw = (\alpha + i)w$ and

$$\bar{v} = u - iw = (\alpha - i)w,$$

Contradicting lin. indep. of $v, \bar{v}$ (over $\mathbb{C}$).

Similarly, if $w = \beta u$, we also reach a contradiction.

$\Rightarrow \{u, w\}$ are lin. indep.



Rmk In particular, $[u, w]_{2 \times 2}$ is invertible.

To obtain a "real normal form" of A , we proceed as follows.

Have $Av = \lambda v$, $\left(w/ \quad A \in \mathbb{R}^{2 \times 2}, \quad v = u + iw, \quad \lambda = a + ib, \right.$
 $\left. a, b \in \mathbb{R}, \quad b \neq 0, \quad u, w \in \mathbb{R}^2. \right)$

$$\Rightarrow \quad A(u + iw) = (a + ib)(u + iw) = au - bw + i(aw + bu)$$

||

$$Au + iAw$$

Equating real, imag parts $\Rightarrow$

$$Au = au - bw$$

$$Aw = bu + aw$$

$$\Rightarrow \quad A \cdot [u, w] = [Au, Aw] = [au - bw, bu + aw]$$

$$= [u, w] \begin{bmatrix} a & b \\ -b & a \end{bmatrix}$$

Since $T := [u, w]$ is invertible by the corollary above, get

$$A = T \begin{bmatrix} a & b \\ -b & a \end{bmatrix} T^{-1}$$

Real normal form of A

$$\Rightarrow e^{At} = T \exp\left(\begin{bmatrix} a & b \\ -b & a \end{bmatrix} t\right) T^{-1} = e^{at} T \begin{bmatrix} \cos(bt) & \sin(bt) \\ -\sin(bt) & \cos(bt) \end{bmatrix} T^{-1}.$$

Ex

Consider $A = \begin{bmatrix} 0 & -2 \\ 1 & 2 \end{bmatrix}.$

$$\det(A - \lambda I) = \det \begin{bmatrix} -\lambda & -2 \\ 1 & 2-\lambda \end{bmatrix} = \lambda(\lambda-2)+2 = (\lambda-1)^2+1 \Rightarrow \lambda = 1+i, \bar{\lambda} = 1-i$$

are the eigenvalues
of A .

It can be shown that $v = \begin{bmatrix} -1+i \\ 1 \end{bmatrix}$ is an e-vector for λ .

$$\lambda = \underbrace{1 + i}_{a} \underbrace{1}_{b}, \quad v = \underbrace{\begin{bmatrix} -1 \\ 1 \end{bmatrix}}_u + i \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_w, \quad \text{so if } T := [u, w] = \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix},$$

$$\text{then } T^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \quad \text{and hence } e^{At} = e^t T \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix} T^{-1}.$$

Real normal form for diagonalizable $A \in \mathbb{R}^{n \times n}$

A has n lin. indep (possibly complex) e -vectors

$$\underbrace{\{v_1, v_2, \dots, v_k\}}_{\text{for real } e\text{-values}}, \quad \underbrace{\{u_{k+1} + i w_{k+1}, u_{k+1} - i w_{k+1}, \dots, u_r + i w_r, u_r - i w_r\}}_{\text{for complex } e\text{-values}}$$

$$\text{Define } P \in \mathbb{R}^{n \times n} \text{ by } P := [v_1, v_2, \dots, v_k, u_{k+1}, w_{k+1}, \dots, u_r, w_r]$$

and note that P is invertible by an extension of our 2×2 reasoning.

Define a block diagonal matrix $D \in \mathbb{R}^{n \times n}$ by

$$D = \begin{bmatrix} \lambda_1 & & & & & \\ & \lambda_2 & & & & \\ & & \ddots & & & \\ & & & \lambda_k & & \\ & & & & B_{k+1} & \\ & & & & & B_{k+2} & \ddots & \\ & & & & & & & B_r \end{bmatrix}$$

Real normal form of A

where $\lambda_1, \dots, \lambda_k$ are the real e-values, and

$$B_j = \begin{bmatrix} a_j & b_j \\ -b_j & a_j \end{bmatrix} \in \mathbb{R}^{2 \times 2}$$

w/ $a_j + ib_j$ the non-real e-values,

Then $A = P D P^{-1}$, so $e^{At} = P e^{Dt} P^{-1}$

where

$$e^{Dt} = \begin{bmatrix} e^{\lambda_1 t} & & & & \\ & e^{\lambda_2 t} & & & \\ & & \ddots & & \\ & & & e^{\lambda_k t} & \\ & & & & e^{B_{k+1} t} \\ & & & & & \ddots & \\ & & & & & & e^{B_r t} \end{bmatrix}$$

Non-diagonalizable case for $A \in \mathbb{R}^{n \times n}$

Let $\lambda_1, \dots, \lambda_p \in \mathbb{R}$ be the real e-values of A and $a_i \pm ib_i, \dots, a_q \pm ib_q \in \mathbb{C}$ be the non-real e-values of A (w/ $a_k, b_k \in \mathbb{R}, b_k \neq 0 \forall k$), both without repetition,

Real Jordan/canonical/normal form thm

$\exists$ an invertible matrix $P \in \mathbb{R}^{n \times n}$ and a block diagonal matrix $J \in \mathbb{R}^{n \times n}$ s.t. $A = PJP^{-1}$ and

$$J = \begin{bmatrix} J_1(\lambda_1) & & & \\ & \ddots & & \\ & & J_p(\lambda_p) & \\ & & & J_1(a_1, b_1) \\ & & & & \ddots \\ & & & & & J_q(a_q, b_q) \end{bmatrix},$$

⋮

where each Jordan λ_k -block $J_k(\lambda_k)$ is block-diagonal w/ diagonal blocks given by elementary λ_k -blocks of the form

$$\begin{bmatrix} \lambda_k & & \\ & \ddots & \\ & & \lambda_k \end{bmatrix}$$

and where each real Jordan $(a_k + ib_k)$ -block $J_k(a_k, b_k)$ is block diagonal w/ diagonal blocks given by real elementary $(a_k + ib_k)$ -blocks

$$\begin{bmatrix} Q_k I_2 & & \\ & \ddots & \\ & & Q_k I_2 \end{bmatrix}$$

$$\text{w/ } Q_k = \begin{bmatrix} a_k & b_k \\ -b_k & a_k \end{bmatrix},$$

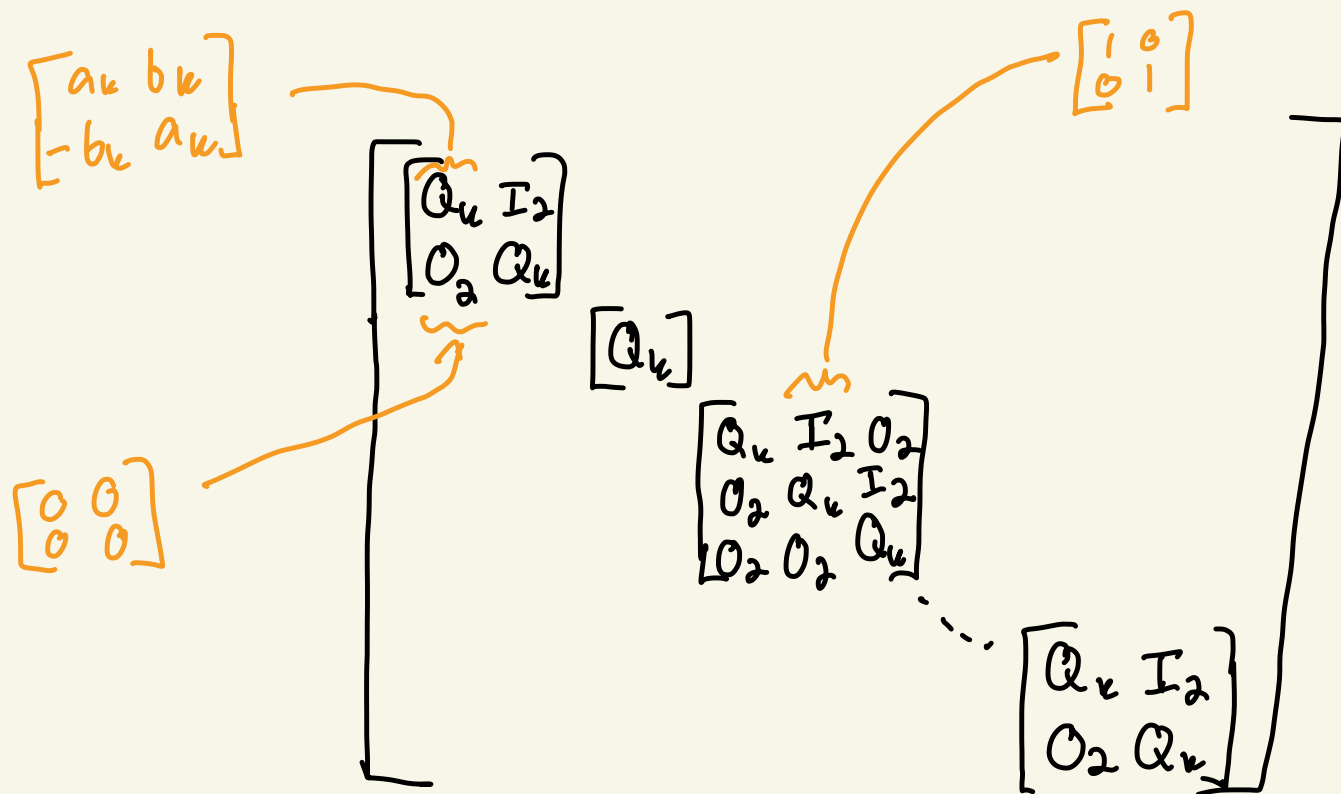
$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

For example, a Jordan λ_k -block might look like

$$\begin{bmatrix} \begin{bmatrix} \lambda_k & 1 & 0 \\ 0 & \lambda_k & 1 \\ 0 & 0 & \lambda_k \end{bmatrix} & & \\ & \begin{bmatrix} \lambda_k \end{bmatrix} & \\ & & \begin{bmatrix} \lambda_k & 1 \\ 0 & \lambda_k \end{bmatrix} & \dots \\ & & & \begin{bmatrix} \lambda_k & 1 & 0 & 0 \\ 0 & \lambda_k & 1 & 0 \\ 0 & 0 & \lambda_k & 1 \\ 0 & 0 & 0 & \lambda_k \end{bmatrix} \end{bmatrix},$$

and similarly a real Jordan $(a_k + ib_k)$ -block might look like

⋮



Remark

To make an analogy, you can think of a real Jordan form as made up of "molecules" given by Jordan λ -blocks and real Jordan $(a+ib)$ -blocks, and these "molecules" are in turn made up of "atoms" given by elementary λ -blocks and real elementary $(a+ib)$ -blocks. In other words, **real Jordan forms are block-diagonal matrices whose diagonal blocks are again block-diagonal matrices.**

